

Problem 1: Choose a start with 6 counters. Lay every pile as a row of counters, longest row on top, like the two pictures below. Write your start on the Start line as numbers, biggest pile first: 4-2, not 2-4. A move: take one counter from every pile, even from a pile of one, and put all the counters you took together as one new pile. Make a move and write the new picture on line 1. Keep going until a picture gives itself back. Count the moves from the start to that picture. Play at least three games from different starts.



Game A		Game B		Game C		Game D	
Start		Start		Start		Start	
1		1		1		1	
2		2		2		2	
3		3		3		3	
4		4		4		4	
5		5		5		5	
6		6		6		6	
7		7		7		7	
8		8		8		8	

Moves to the picture that gives itself back: A _____ B _____ C _____ D _____

The picture that gives itself back: _____ Did everyone at your table reach it? _____

The start that took the most moves at your table: _____, with _____ moves.

Problem 2: Find every picture that can be made with 6 counters. Go in order: first the pictures whose biggest pile is 6, then biggest pile 5, then 4, 3, 2 and 1. Write each picture once in the middle column, biggest pile first. Count how many you found.

Problem 3: For each picture in your list, build it, make one move, and write the picture one move later in the right column.

biggest pile	picture	one move later
6		
5		
4		
3		
2		
1		

Number of pictures with 6 counters: _____

Problem 4: Draw the map of all pictures with 6 counters in the box. Write every picture from your list inside a small circle. Draw an arrow from each picture to the picture one move later. Put the picture that gives itself back near the bottom, so that the arrows flow down. Then answer (a), (b) and (c) under the box.

(a) The start farthest from the bottom picture is _____, and it takes _____ moves.

(b) Pictures that have no arrow pointing into them: _____

(c) A picture that two arrows point into: _____

Problem 5: Now use 4 counters. List every picture with 4 counters and, beside each one, the picture one move later. Draw the map in the box under the list. Then do the same with 5 counters. In each map, circle the pictures that the game goes through again and again, and write that loop as a chain of arrows on the lines at the bottom.

4 counters

picture	one move later

5 counters

picture	one move later

Loop with 4 counters: _____

Loop with 5 counters: _____

Is there a picture with 4 counters that gives itself back? _____ With 5 counters? _____ Tell the adult how your list shows it.

Problem 6: A picture is stuck if one move gives the same picture back. (a) For each picture in the table, build it, make one move, write what you get, and write yes or no. (b) Invent three pictures that you think are stuck and test them in the last three rows. (c) Count the counters in each stuck picture you found and write the counts on line (c), smallest first. (d) Build the next stuck picture, one row bigger than the biggest one you found. Write it and its count on line (d). (e) Tell the adult what happens to each pile of 3-2-1 in one move, pile by pile, and why the same thing happens to 4-3-2-1.

picture	one move later	stuck?
2-1		
3-2-1		
4-3-2-1		
2-2		
4-2		
3-3		
5-4-3-2-1		
6-3-1		

(c) Counts of the stuck pictures, smallest first: _____

(d) The next stuck picture: _____ its count: _____

Problem 7: Now use 10 counters. Game A: start with one pile of 10. Game B: start 5-5. Game C: start 2-2-2-2-2. Game D: a start of your own. Write every picture, biggest pile first, and count the moves to the stuck picture. Then look for a start with 10 counters that takes more than 8 moves. Use games E to H on the next page for the search, and write the start you find and its number of moves at the bottom of that page.

Game A: 10		Game B: 5-5		Game C: 2-2-2-2-2		Game D	
Start		Start		Start		Start	
1		1		1		1	
2		2		2		2	
3		3		3		3	
4		4		4		4	
5		5		5		5	
6		6		6		6	
7		7		7		7	
8		8		8		8	
9		9		9		9	
10		10		10		10	
11		11		11		11	
12		12		12		12	
13		13		13		13	

Problem 7, continued: Games E to H: a start of your own in each one, written in its Start box. Count the moves to the stuck picture. Keep going until one start takes more than 8 moves.

Game E		Game F		Game G		Game H	
Start		Start		Start		Start	
1		1		1		1	
2		2		2		2	
3		3		3		3	
4		4		4		4	
5		5		5		5	
6		6		6		6	
7		7		7		7	
8		8		8		8	
9		9		9		9	
10		10		10		10	
11		11		11		11	
12		12		12		12	
13		13		13		13	

Moves: A _____ B _____ C _____ D _____ E _____ F _____ G _____ H _____

A start that takes more than 8 moves: _____ takes _____ moves.

Problem 8: Now use 8 counters. (a) Game A: start with one pile of 8. Game B: start 2-2-2-2. Write every picture until one repeats. (b) In each game, circle the pictures that the game goes through again and again, and write that loop on its line. (c) Play from 5-3, from 4-4 and from 6-1-1, and write which loop each start reaches in the small table. Add starts of your own in the empty rows. (d) Draw both loops in the box at the bottom: the pictures of loop A in a ring with arrows between them, and the pictures of loop B in another ring.

Game A: 8	
Start	
1	
2	
3	
4	
5	
6	
7	
8	
9	

Game B: 2-2-2-2	
Start	
1	
2	
3	
4	
5	
6	
7	
8	
9	

start	reaches loop
5-3	
4-4	
6-1-1	

Loop A: _____

Loop B: _____

Are loop A and loop B the same loop? _____ Pictures in loop A: _____ in loop B: _____

Problem 9: Now use 7 counters. Start with one pile of 7. Write every picture until one repeats, and circle the loop. Draw each loop picture as rows of dots in the boxes, in order. In each drawing, find a 3-2-1 stairs, color its six dots, and put a ring around the dot that is left over. Then look at where the ring is in box 1, box 2, box 3 and box 4.

Start: 7	
Start	
1	
2	
3	
4	
5	
6	
7	

1	2
3	4

The loop with 7 counters: _____

Which row is the ringed dot in? box 1 ____ box 2 ____ box 3 ____ box 4 ____

Problem 10: Now use 9 counters. Start with one pile of 9. Write every picture until one repeats, and circle the loop. Draw the loop pictures in the boxes, in order. In each drawing, find a 3-2-1 stairs, color its six dots, and ring every dot that is left over. Watch how the ringed dots move from box to box.

Start: 9	
Start	
1	
2	
3	
4	
5	
6	
7	
8	
9	

1	2
3	4

Problem 11: (a) Build each picture in the table. Before you make the move, predict two things and write them down: how many counters the new pile will have, and how many piles there will be after the move. (b) Then make the move, write the picture you get, and check your predictions. (c) Add two pictures of your own in the last rows. (d) Tell the adult how you can know the size of the new pile before you move, and how you can know the number of piles after the move.

picture	new pile will have	piles after the move	one move later
5-3-2			
4-4-1-1			
6-1-1-1			
2-2-2-2-2			
3-3-2-1-1			
7-1			
3-2-2-1			
5-1-1-1-1-1			

Problem 12: Look at the pictures on your map from Problem 4 that no arrow points into. (a) Pretend six piles of 1 had just been made by a move. One of the piles is the new pile. Work out what the picture before the move must have been, write it, make the move from it, and write what you get. (b) Now pretend 2-2-1-1 had just been made. Pick one of its piles to be the new pile and give one counter back from it to every other pile. Try each pile. Write what goes wrong. (c) Do the same for 2-1-1-1-1.

(a)

(b)

(c)
